

PROJECT PROPOSAL

Salary Prediction and Income Analysis

Course: Fundamental of Data Science

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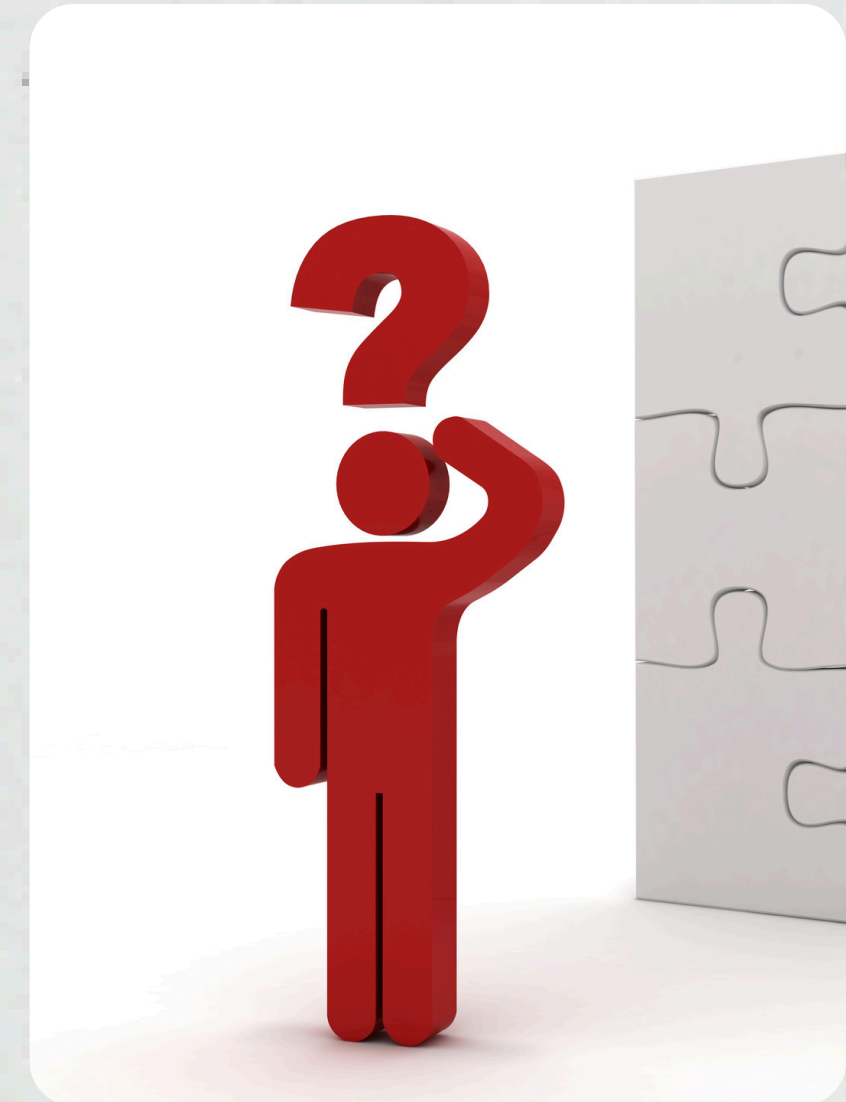


INTRODUCTION

This proposal present a complete data science study that analyzes income determinants and predict salary levels using the Adult income dataset from the UCI Machine Learning Repository. [1]

PROBLEM

Many students do not clearly understand how education, skills, experience, and job roles influence expected salaries. This lack of transparency can lead to poor career decisions, unrealistic salary expectations, and a mismatch between acquired skills and market demand.



ref: <https://www.jstor.org/stable/1925454?origin=JSTOR-pdf&seq=1>

LITURER REVIEW

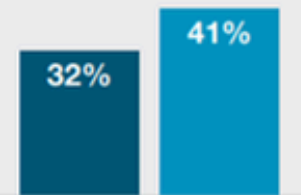
CONFIDENT WITH THEIR SKILL AND KNOWLEDGE

- Overall, 34% and 36% of current students believe they will graduate with the skills and knowledge they need to be successful in the job market. [3]
- A majority of students (53%) believe their major will lead to a good job.[3]

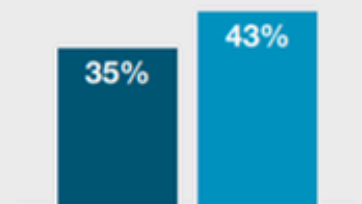
Workforce preparation, by student age

■ Students younger than 24 ■ Students aged 24 or older

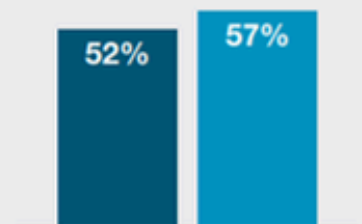
I am confident I will graduate with the knowledge and skills I need to be successful in the job market.



I am confident I will graduate with the knowledge and skills I need to be successful in the workplace.



I am confident my major field of study/studies will lead to a good job.



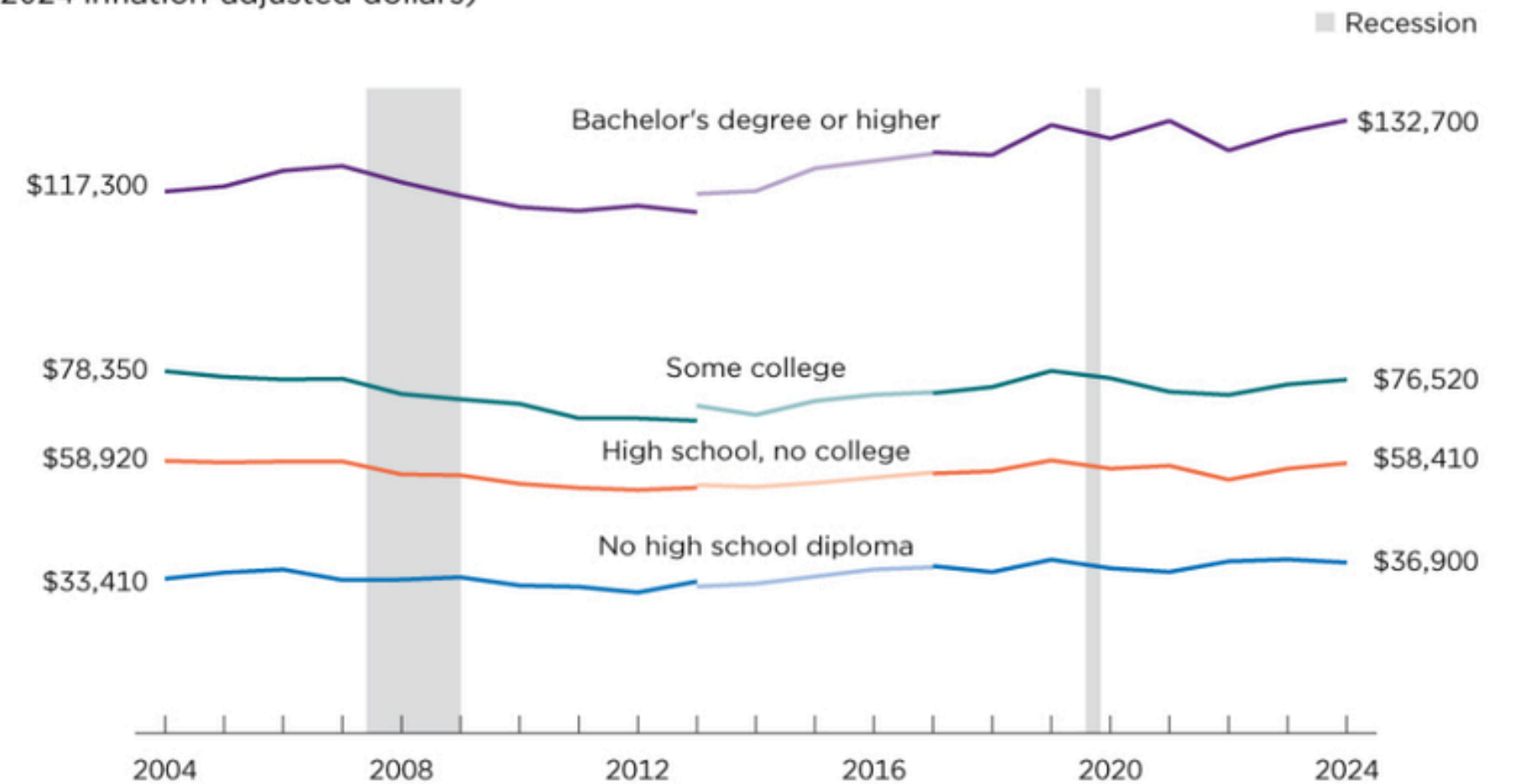
Strada Education Network [3]

LITURER REVIEW

EDUCATION IMPACTED INCOME

- Estimated median household income of householders with only a high school degree didn't change significantly from 2004 (\$58,410) to 2024 (\$58,920). [2]

Figure 1.
**U.S. Median Household Income by Householder's Educational Attainment:
2004–2024**
(2024 inflation-adjusted dollars)



Z. Scherer and M. D. King [2]



SOLUTION

Develop a salary prediction system using data analytics and machine learning to estimate expected salaries based on relevant factors such as education, skills, experience, and job roles. This system can guide individuals in making informed career decisions.



FOCUS

The project focuses on:

- Data understanding
- Cleaning
- Exploratory analysis
- Statistical insight
- Feature engineering
- Machine learning modeling.

*The scope ends at model evaluation and does not include system deployment.

PROJECT'S OBJECTIVE

- Understand income distribution patterns.
- Study how education level influences salary
- Identify key factors affecting higher income
- Build and compare predictive models
- Generate interpretable insight for decision making



DATASET METADATA

Attribute	Type	Description
age	Numerical	Age of individual
workclass	Categorical	Employee type
fnlwgt	Numerical	Sampling weight (removed)
education	Categorical	Education level
education num	Numerical	Education encoded numerically
marital status	Categorical	Marital Status
occupation	Categorical	Job role
relationship	Categorical	Family relationship
race	Categorical	Race group
sex	Categorical	Gender
capital-gain	Numerical	Investment gains
capital-loss	Numerical	Investment loss
hour-per-week	Numerical	Working hours per week
native-country	Categorical	Country of origin
income	Categorical	Salary class ($\leq 50K$ or $> 50K$)

ref: <https://archive.ics.uci.edu/dataset/20/census+income>

STEP 1

Data Understanding

- Inspect dataset size, structure, and data types.
- Identify missing values and anomalies
- Understand class imbalance
- Summarize numerical and categorical distributions

STEP 2

Data Cleaning

- Remove irrelevant attributes such as fnlwgt
- Handle missing or unknown values
- Fix inconsistent categories
- Remove duplicates and outliers

STEP 3

Exploration Data Analysis

- Analyze salary distribution
- Compare education level vs income
- Study occupation and workclass trends
- Visualize age and hours-per-week impacts
- Generate correlation heatmaps and boxplots

STEP 4

Statistical Analysis & Insight Extraction

- Group-by comparisons for income trends
- Hypothesis testing for education impact
- Measure correlation for education impact
- Interpret which attributes influence salary most

STEP 5

Feature Engineering

- Encode categorical variables using One-Hot or Label Encoding
- Create education rank scores
- Create overtime indicator (hours > 40)
- Group rare categories
- Scale numeric features

STEP 6

Machine Learning Modeling

- Train Logistic Regression as baseline
- Train Decision Tree for interpretability
- Train Random Forest and Gradient Boosting for higher accuracy
- Perform train-test split and cross-validation

STEP 7

Model Evaluation

- Evaluate using accuracy, precision, recall, and F1-score
- Generate confusion matrix and ROC curve
- Compare models and select best performer
- Analyze feature importance

STEP 8

Reporting & Conclusion

- Summarize findings and insights
- Discuss education impact on salary
- Provide recommendations
- Documentation limitations and future improvements

OUT SCOPE

- API Development
- Web application
- Deployment

REFERENCE

- [1] "Census Income," UCI Machine Learning Repository, Feb. 23, 2026. [[Link](#)].
- [2] Z. Scherer and M. D. King, "How Education Impacted Income and Earnings From 2004 to 2024," *U.S. Census Bureau*, Sept. 09, 2025. [[Link](#)].
- [3] Strada Education Network, *Crisis of Confidence: Strada Student Perspectives Survey 2018*, 2018. [[Link](#)].
- [4] M. Xu, "Salary prediction using machine learning," *Scholarly Review Journal*, Jun. 2025, doi:10.70121/001c.139043. [[Link](#)].



THANK YOU